MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, COAXIAL, SIZE 12 (FOR MIL-C-38999 SERIES I, III, AND IV CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.

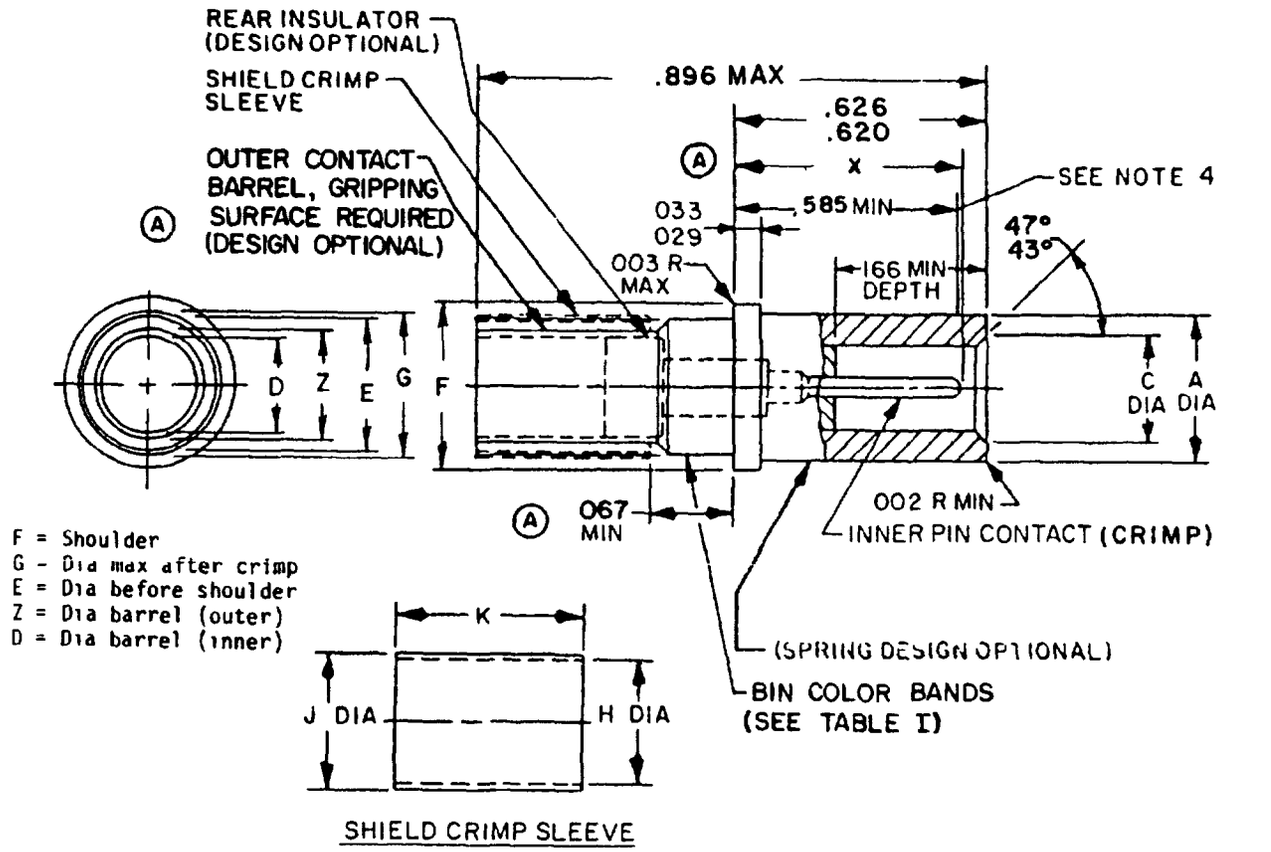


FIGURE 1 Socket contacts

(A) denotes change

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Bin code	A dia	C dia	D dia	E dia	F dia	G dia max	H dia min	J dia max	K	L dia min
559	.160	.123	.069	.151	.182	.156	.127	.169	.180	.022
	.158	.118	.066	.148	.179				.170	

Bin code	M dia max	N min	P	R	S	T ref	U	V dia	W dia	X	Z dia max
	.034	.070	.056	.228	.155	.373	.603	.0205	.026	.613	.110
559			.036	.226	.151		.597	.0195	.024	.603	

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0 08	.033	0.84	.110	2.79	.158	4.01	.226	5.74	.620	15.75
.013	0 33	.034	0.86	.118	3.00	161	4.09	.228	5.79	622	15.80
.0195	0 495	036	0.91	.123	3.12	.166	4.22	373	9.47	626	15.90
.0205	0.521	.056	1.42	.127	3.23	169	4.29	.585	14.86	.896	22.76
022	0.56	.063	1.60	.148	3.76	.170	4.32	.597	15.16		
024	0 61	064	1.63	.151	3.84	.179	4.55	.600	15.24		
026	0.56	.066	1.68	.155	3 94	180	4 57	603	15 32		
029	0.74	070	1.78	.156	3.96	.182	4.62	613	15.57		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the inner contact spring Provision for clearance hole shall be provided.
5 Crimp deformation The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed G diameter.

FIGURE 1. Socket contacts - Continued.

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REQUIREMENTS:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Mating contacts: MIL-C-39029/102.

Tools. See table II.

Contact resistance. See table III.

Test current:

Inner contact: 1.0 ampere.

Outer contact: 12.0 amperes

Low signal level contact resistance (inner contact only): See table IV.

Nominal impedance: 50 ohms.

TABLE I. Design characteristics.

Bin code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
559	Green	Green	White	M17/113-RG316 M17/094-RG179	12	D	B

TABLE II. Tools.

Bin code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Ⓐ Die		
559	N/A	N/A	M22520/5-01	M22520/5-03	M81969/8-09 or M81969/14-04	M81969/8-10 or M81969/14-04

TABLE III. Contact resistance.

Bin code	Cable accommodated	Maximum voltage drop (millivolts)						Maximum average voltage drop
		25°C + 3°C -0°C		25°C +3°C -0°C 1/		200°C +3°C -0°C		
		Inner contact	Outer contact	Inner contact	Outer contact	Inner contact	Outer contact	
559	M17/113-RG316	55	75	66	90	94	128	N/A
	M17/094-RG179	120	70	144	84	204	119	N/A

1/ After conditioning.

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TABLE IV. Low signal level contact resistance (inner contact only) and tensile strength.

Bin code	Cable accommodated	Maximum contact resistance (milliohms)		Tensile load (pounds) (minimum)	
		Initial	After conditioning	Inner contact	Outer contact
559	M17/113-RG316	55	66	10.0	15.0
	M17/094-RG179	120	144	3.5	15.0

TABLE V. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0950 +.0002 -0000	N/A	N/A	30	36	N/A
.0930 + 0000 -.0002	3.0	2.5	N/A	N/A	N/A

Voltage standing wave ratio (VSWR) (applicable to M17/113-RG316 cable only).

(A)

(A) a. The VSWR of this socket contact mated with a MIL-C-39029/102 pin contact shall not exceed 1.20 + .04F (F in GHz) from 500 MHz to 3 GHz under all of the following conditions:

(1) Pin and socket contacts fully mated.
(A) (2) Pin and socket contacts .050 ±.005 inches from full engagement.
(A) (3) Pin and socket contacts .100 ±.005 inches from full engagement.

(A) b Swept frequency VSWR test setup
Mated pair of contacts shall be tested using a Hewlett-Packard 8510 Vector Network Analyzer as follows

(1) Use the Hewlett-Packard 8510 calibrated at the output port or at a connector saver connected to the output port
(2) Sweep-Ramp mode, 401 points, 500 MHz to 3 0 GHz.
(3) Make up one RG316 cable assembly (approximately 6 inches long) with a MIL-C-39029/102 on one end and an SMA connector on the other end
(4) Make up another RG316 cable assembly (approximately one foot long) with a MIL-C-39029/103 on one end and an SMA connector on the other end
(5) Terminate the SMA connector on the longer assembly with a 50 ohm load and fully mate the MIL-C-39029/102 and MIL-C-39029/103 contacts
(6) Connect the other SMA connector to the APC-3 5 port which had been calibrated
(7) Change the display to time domain, band pass
(8) Set the gate center at the coaxial contact interface and the gate width at 2 25 nanoseconds
(9) Turn the gate on and return to the frequency domain
(10) Plot the connector pair VSWR (S_{11})
(11) Verify that the displayed VSWR (S_{11}) complies with the VSWR limits specified for the mated pair of contacts
(12) Repeat (5) through (11) with pin and socket contacts 050 ± 005 and 100 ± 005 inches from full engagement

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Insertion loss (applicable to M17/113-RG316 cable only): dB maximum = .11 x $\sqrt{F}$ (A) (F in GHz). When measured in accordance with MIL-C-39012 at 3 GHz, the insertion loss shall not exceed .20 dB.

Contact engagement and separation forces (inner socket contact only): The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in table V, and surface roughness shall not exceed 3 microinches. Provision for clearance hole shall be provided (see table V).

Dielectric withstanding voltage (applied between inner and outer contact):

Test voltage.

At sea level - 1000 V rms

At 50,000 feet - 250 V rms

(A) Tensile strength (inner and outer contact): See table IV.

Vibration. Test condition D, in accordance with method 204 of MIL-STD-202. Except the vibration frequency shall be sinusoidal.

Random vibration: Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344. The following details shall apply.

a. Test condition V. Using the vibration envelope.

b. Vibration to be conducted at standard test conditions.

c. Duration shall be 8 hours in the longitudinal and 8 hours in the perpendicular direction for a total of 16 hours.

High impact shock: All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock. Contacts shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated contacts shall be subjected to the test specified in MIL-C-901, grade A with (A) the following modifications and additions. Mounting fixture shall be in accordance with MIL-C-901, light weight. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. Coupling Coupled by normal mounting means.

b. Test condition A. The wire bundles shall be provided with straight, open frame, relief accessory hardware.

Thermal shock: Test condition B, in accordance with method 107 of MIL-STD-202. Except high temperature dependent upon material.

Shipping. Figure 1 illustrates final assembly, parts are to be shipped unassembled in a suitable container. Manufacturers recommended assembly instructions are to be shipped with the unit package.

QPL evaluating activity Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

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Military part number and supersession data See table VI.

TABLE VI. Military part number and supersession data.

Bin code	Military part number	Supersedes part number
559	M39029/103-559	N/A

Custodians.
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR, AV
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

User activity
Air Force - 19

Preparing activity.
Air Force - 85

Agent
DLA - ES

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